Package ‘BayesRank’

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Type Package

Title Bayesian rank-based isotonic regression for the analysis of Caenorhabditis elegans toxicological assays

Version 0.1.0

Description Implements a Bayesian rank-likelihood model for ordinal regression under monotonicity constraints, developed to analyze the Caenorhabditis elegans neurotoxicity assay from Bergemann et al. (2026) 'Progeny effects of rotenone exposure depend on parental toxicity' (Toxicological Sciences). Reproduces the analysis from Presman, Anceschi, Huayta, Meyer, and Herring (2026+) 'Order-Restricted Bayesian Ordinal Regression for the Modeling of Neuron Degeneration in Caenorhabditis elegans' (forthcoming), extending existing rank-likelihood software to support random effects, parameter constraints, collapsed sampling, and parameter expansion. While the package's code is specific to the target assay design, the implemented modeling framework readily generalizes to a much broader range of rank-based isotonic regression settings.

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patchwork,
viridis,
scales

LinkingTo Rcpp,
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Contents

`fit_BayesRank` *Gibbs Sampler for Bayesian Isotonic Rank Likelihood Model*

Description

Fits a Bayesian rank-based isotonic regression model via Gibbs sampling. Optional analytic marginalization of random effects (after burn-in) and parameter expansion for selected updates are available for improved mixing.

Usage

```
fit_BayesRank(
  y,
  X,
  W,
  Q,
  MC_seed = 0,
  iter_MC = 10000,
  iter_Burn = 5000,
  hyper_params = NULL,
  marg_RE = T,
  PX = T,
  rcpp = T
)
```

Arguments

`y`	Ordinal response vector (values 1:K).
`X`	Design matrix for treatment coefficients.
`W`	Design matrix for group-specific intercepts.
`Q`	Incidence matrix assigning observations to random-effect groups.

`MC_seed`	Random seed. Default 0.
`iter_MC`	Total MCMC iterations (must exceed iter_Burn). Default 10000.
`iter_Burn`	Burn-in iterations. Default 5000.
`hyper_params`	List of hyper-parameters (see `set_hyperparameters`)
`marg_RE`	Marginalize random effects after burn-in? Default TRUE.
`PX`	Use parameter expansion (PX-DA)? Default TRUE.
`rcpp`	Use Rcpp for multivariate truncated normal sampling? (Only if `marg_RE` is TRUE) Default TRUE.

Value

List with posterior draws (including warmup) for

alpha Incremental treatment coefficients.

beta Cumulative treatment effects.

mu Replicate intercepts, excluding the constrained last replicate (sum-to-zero).

eta Random effects.

rho2 Variance of random effects.

Z Latent utilities of the rank likelihood.

The list also includes hyper-parameters and MCMC settings.

References

Presman, Anceschi, Huayta, Meyer, and Herring, (2026+) "Order-Restricted Bayesian Ordinal Regression for the Modeling of Neuron Degeneration in Caenorhabditis elegans"

`get_data` *Load and preprocess C. elegans neuron-level data*

Description

Reads the raw neuron-level CSV, splits the combined `Treatment` field into `Generation`, `Treatment`, and `Retreatment`, constructs a unique worm identifier, converts categorical variables to factors.

Usage

```
get_data()
```

Value

A data frame with columns: `Replicate`, `Treatment`, `NeuronScore`, `UniqueWormID`, `Retreatment`, `Generation`, `logNeuronScore`, `Worm`.

References

Bergemann et al. (2026) 'Progeny effects of rotenone exposure depend on parental toxicity', Toxicological Sciences, Volume 209, Issue 3, March 2026, kfag011

get_data_BayesRank — *Build model matrices for the Bayesian rank-based isotonic regression*

Description

Constructs the response vector and design matrices required by `fit_BayesRank` from preprocessed neuron-level data. `X` encodes a cell-means representation of the parental early Treatment ("Low" and "High vs. Low") crossed with Generation and Retreatment. `Q` is the worm-level random-effect incidence matrix, and `W` is the replicate (batch) design matrix.

Usage

```
get_data_BayesRank(raw_data)
```

Arguments

`raw_data`	Preprocessed data frame, as returned by `get_data`.

Value

A list with elements:

- **y** Ordinal neuron damage score.
- **X** Design matrix of three-way interaction terms (parental early Treatment crossed with Generation and Retreatment
- **W** Replicate design matrix.
- **Q** Worm-level random-effect incidence matrix.

get_data_clmm — *Build model matrix and data frame for the CLMM comparison model*

Description

Constructs the fixed-effect design matrix for the cumulative link mixed model (`clmm`) benchmark. Response variables and the worm-level grouping factor are appended for use with `ordinal::clmm`.

Usage

```
get_data_clmm(raw_data)
```

Arguments

`raw_data`	Preprocessed data frame, as returned by `get_data`.

Value

A data frame with the fixed-effect dummy-coding columns for Replicate membership and interactions as in `Treatment:Retreatment:Generation`. For the three-way interactions, the columns involving the control Treatment group serve as the implicit within-stratum reference, and are thus dropped. Additional info include NeuronScore, logNeuronScore, factNeuronScore, and UniqueWormID.

`neuron_level_data` *C. elegans neuron-level damage scores*

Description

Raw neuron-level damage score data from a C. elegans developmental rotenone exposure assay, recording one row per neuron observed across worms, replicates, generations, treatments, and rechallenge status.

Usage

```
neuron_level_data
```

Format

An object of class `data.frame` with 2099 rows and 4 columns.

Source

Bergemann et al. (2026), Toxicological Sciences, 209(3):kfag011.

`plot_coeff_cumulative` *Plot posterior distributions of cumulative treatment effects*

Description

Plots histograms of posterior draws for the "Low" and "High" cumulative treatment coefficients, faceted by Generation and Rechallenge.

Usage

```
plot_coeff_cumulative(
  draws_mcmc,
  show = T,
  save = F,
  filename = "coeff_cumulative",
  dir = "~/Desktop"
)
```

Arguments

`draws_mcmc`	Postprocessed MCMC output from `postprocess_BayesRank`
`show`	Display the plot to the active graphics device? Default TRUE.
`save`	Save the plot to a PDF file? Default FALSE.
`filename`	Output file name (without extension). Default 'coeff_cumulative'.
`dir`	Output directory. Default '~/Desktop'.

Value

Invisibly returns the ggplot object.

plot_coeff_incremental

Plot posterior distributions of incremental treatment effects

Description

Plots histograms of posterior draws for the "Low" and "High vs. Low" incremental treatment coefficients, faceted by Generation and Rechallenge.

Usage

```
plot_coeff_incremental(
  draws_mcmc,
  show = T,
  save = F,
  filename = "coeff_incremental",
  dir = "~/Desktop"
)
```

Arguments

draws_mcmc	Postprocessed MCMC output from postprocess_BayesRank
show	Display the plot to the active graphics device? Default TRUE.
save	Save the plot to a PDF file? Default FALSE.
filename	Output file name (without extension). Default 'coeff_incremental'.
dir	Output directory. Default '~/Desktop'.

Value

Invisibly returns the ggplot object.

plot_latent_scores

Plot category-wise posterior distributions of latent scores

Description

Plots overlapping histograms of the latent utilities, groupe and colored by observed ordinal category, to visualize separation between adjacent groups.

Usage

```
plot_latent_scores(
  draws_mcmc,
  show = T,
  save = F,
  filename = "hist_Z_scores",
  dir = "~/Desktop"
)
```

Arguments

`draws_mcmc`	Postprocessed MCMC output from `postprocess_BayesRank`
`show`	Display the plot to the active graphics device? Default TRUE.
`save`	Save the plot to a PDF file? Default FALSE.
`filename`	Output file name (without extension). Default 'hist_Z_scores'.
`dir`	Output directory. Default '~/Desktop'.

Value

Invisibly returns the ggplot object.

`plot_predictions` *Plot a confusion matrix of observed vs. predicted neuron scores*

Description

Plots a heatmap-style confusion matrix comparing observed and predicted ordinal categories

Usage

```
plot_predictions(
  Observed,
  Predicted,
  method = NULL,
  colors = "mako",
  show = T,
  save = F,
  filename = "confusion_matrix",
  dir = "~/Desktop"
)
```

Arguments

`Observed`	Vector of observed ordinal categories.
`Predicted`	Vector of predicted ordinal categories.
`method`	Optional label identifying the prediction method, appended to the plot title. Default NULL.
`colors`	Viridis color palette option used for cell fill. Default 'mako'.
`show`	Display the plot to the active graphics device? Default TRUE.
`save`	Save the plot to a PDF file? Default FALSE.
`filename`	Output file name (without extension). Default 'confusion_matrix'.
`dir`	Output directory. Default '~/Desktop'.

Value

Invisibly returns the ggplot object.

plot_random_effects *Plot posterior medians of random effects against category boundaries*

Description

Plots an histogram of posterior medians of the worm-level random effects. Overlays vertical reference lines and colored markers, showing the estimated boundaries on the latent scale among ordinal categories.

Usage

```
plot_random_effects(
  draws_mcmc,
  show = T,
  save = F,
  filename = "hist_RE_medians",
  dir = "~/Desktop"
)
```

Arguments

draws_mcmc	Postprocessed MCMC output from postprocess_BayesRank
show	Display the plot to the active graphics device? Default TRUE.
save	Save the plot to a PDF file? Default FALSE.
filename	Output file name (without extension). Default 'hist_RE_medians'.
dir	Output directory. Default '~/Desktop'.

Value

Invisibly returns the ggplot object.

plot_replicates_ICC *Plot posterior distributions of batch effects and intraclass correlation*

Description

Plots posterior distributions of replicate-level intercept shifts alongside the intraclass correlation (ICC), combined into a single side-by-side figure.

Usage

```
plot_replicates_ICC(
  draws_mcmc,
  show = T,
  save = F,
  filename = "replicates_icc",
  dir = "~/Desktop"
)
```

Arguments

draws_mcmc	Postprocessed MCMC output from postprocess_BayesRank
show	Display the plot to the active graphics device? Default TRUE.
save	Save the plot to a PDF file? Default FALSE.
filename	Output file name (without extension). Default 'replicates_icc'.
dir	Output directory. Default '~/Desktop'.

Value

Invisibly returns the ggplot object.

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postprocess_BayesRank *Post-process Gibbs sampler output after fitting on the C. elegans isotonic data*

Description

Discards burn-in draws and derives quantities of interest from the raw MCMC output of fit_BayesRank, intended for use by downstream plotting functions.

Usage

```
postprocess_BayesRank(fit_mcmc, input_data)
```

Arguments

fit_mcmc	Output list from fit_BayesRank.
input_data	List containing the original model inputs.

Value

A list with elements:

alpha Post-burn-in incremental treatment coefficients. ("Low" and "High vs. Low").

beta Post-burn-in cumulative treatment effects (with "High vs. Low" columns converted to "High")

mu Post-burn-in replicate intercepts, including the constrained last replicate (sum-to-zero).

eta Post-burn-in random effects.

median_RE Posterior median of each random effect.

icc Posterior draws of the intraclass correlation, rho2/(1+rho2).

Zy Post-burn-in latent utilities, split by ordinal category.

Zmin Per-draw minimum latent utility within each category.

Zmax Per-draw maximum latent utility within each category.

Note

Assumes the specific column-name convention ("Low"/"High vs. Low, , <Un-/Rechallenged>") produced by the paper's design matrix (i.e. not a general-purpose postprocessing routine).

`predict_BayesRank` *Predict ordinal damage scores from posterior draws of the Bayesian rank-based isotonic model*

Description

Computes posterior predictive draws of the ordinal response category by combining posterior draws of treatment effects, replicate intercepts, and random effects into a linear predictor. The resulting continuous score is then mapped it to an ordinal category via the posterior estimates of the cutpoints, estimated from the inferred separation between latent utility.

Usage

```
predict_BayesRank(X, W, Q, vals_mcmc)
```

Arguments

`X`	Design matrix for treatment coefficients.
`W`	Design matrix for group-specific intercepts.
`Q`	Incidence matrix assigning observations to random-effect groups.
`vals_mcmc`	Post-processed MCMC output from `postprocess_bayes`

Value

A list with elements:

y_mcmc Matrix of predicted categories, one row per observation and one column per post-burn-in MCMC draw.

y_median Vector of median predicted category per observation, taken across MCMC draws.

`predict_clmm` *Predict ordinal class from a fitted cumulative link mixed model*

Description

Computes predicted ordinal categories from a fitted `clmm` model by combining the fixed-effect linear predictor with subject-specific random-effect, then mapping the result to a category via the model's estimated cutpoints.

Usage

```
predict_clmm(fit_clmm, df_ord)
```

Arguments

`fit_clmm`	A fitted `clmm` object (from package `ordinal`)
`df_ord`	Data used to fit `clmm` frame (or one with the same format

Value

A factor of predicted categories, with levels matching `fit_clmm$y.levels`.

`set_hyperparameters` *Check and fill defaults for Gibbs sampler hyperparameters*

Description

Validates a list of hyperparameters for `gibbs_sampler`, filling in default values for any missing elements.

Usage

```
set_hyperparameters(hyper_params = NULL)
```

Arguments

`hyper_params` List of hyperparameters, with elements:

- **w0** Prior SD for normla prior on the group-specific intercepts. Default 1.
- **pi0** Prior spike probability for treatment coefficients. Default 0.5.
- **mu0** Prior mean for the gaussian slab component of treatment coefficients. Default 0.
- **lambda0** Prior SD for the gaussian slab component of treatment coefficients. Default 1.
- **r_shape** Shape hyperparameter for the inverse-gamma prior on the random-effect variance (rho2). Default 2.
- **r_rate** Rate hyperparameter for the inverse-gamma prior on the random-effect variance (rho2). Default 2.
- **iter_MH** Inner sweeps for multivariate truncated normal Gibbs sampling. Default 10.
- **rho2_GG** Initial/center value for Griddy-Gibbs on rho2. Default 2.
- **rho2_max** Upper bound for Griddy-Gibbs on rho2. Default 5.
- **n_grid_rho2** Number of grid points for Griddy-Gibbs on rho2. Default 100.

May be partially specified or NULL; missing elements are filled with defaults.

Value

A complete list of hyperparameters.

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